CHE 201 Spring 2020 Midterm 2

Roxanna Mendoza

TOTAL POINTS

69 / 100

QUESTION 1

Question 1 25 pts

1.1 A 12 / 21

- **- 0 pts** Correct
- ✓ **- 3 pts first one is incorrect**
- **- 3 pts** second one is incorrect
- **- 3 pts** third one is incorrect
- ✓ **- 3 pts fourth one is incorrect**
- **- 3 pts** fifth one is incorrect
- **- 3 pts** sixth one is incorrect
- ✓ **- 3 pts seventh one is incorrect**

1.2 B 4 / 4

- ✓ **- 0 pts Correct**
- **- 4 pts** not done or incorrect

QUESTION 2

Question 2 40 pts

2.1 a 10 / 10

- ✓ **- 0 pts Correct**
- **- 3 pts** Qc not labeled or wrong direction
- **- 3 pts** Qh not labeled or wrong direction
- **- 3 pts** work not labeled or wrong direction

2.2 b 5 / 5

- ✓ **- 0 pts Correct**
- **- 5 pts** incorrect or not done
- **- 2 pts** only one calculated
- **- 2 pts** partially done/not calculated for year
- **- 2 pts** without cooling is incorrect

2.3 C 0 / 5

- **- 0 pts** Correct
- ✓ **- 5 pts not done/incorrect**
- **- 1 pts** calculation error

2.4 d 2 / 5

- **- 0 pts** Correct
- **- 5 pts** incorrect/not done
- ✓ **- 3 pts Th is taken as 90C**

2.5 e 10 / 10

- ✓ **- 0 pts Correct**
- **- 2 pts** calculation error
- **- 10 pts** not done/incorrect
- **- 5 pts** only one calculated
- **- 2 pts** unit conversion

2.6 f 0 / 5

- **- 0 pts** Correct
- ✓ **- 5 pts incorrect /reverse formula/ not done**
- **- 1 pts** reversible/irreversible?
- **- 1 pts** calculation error

QUESTION 3

Question 3 35 pts

3.1 a 2 / 5

- **- 0 pts** Correct
- ✓ **- 3 pts two outlets are not shown**
- **- 2 pts** inlet is not shown
- **- 1 pts** boundary is not labeled

3.2 b 5 / 5

- ✓ **- 0 pts Correct**
- **- 2 pts** open system not stated
- **- 2 pts** steady state is not stated
- **- 1 pts** delta PE is zero

3.3 C 2 / 5

- **0 pts** Correct

✓ - **5 pts** **incorrect**

- **2 pts** work can not be canceled

\+ **2** **Point adjustment**

3.4 d 10 / 10

✓ - **0 pts** **Correct**

- **2 pts** h2 is wrong

- **2 pts** KE term is not divided by 1000 (unit conversion)

- **10 pts** incorrect or not done

- **2 pts** not multiplied with mass outlets

- **1 pts** work sign is wrong

- **2 pts** work is not added

- **2 pts** work unit is not converted

- **2 pts** mass is wrong

- **2 pts** h1 is wrong

- **2 pts** h3 is wrong

- **0 pts** Click here to replace this description.

calc error

3.5 e 5 / 5

✓ - **0 pts** **Correct**

- **5 pts** incorrect/not done

- **2 pts** specific volume is wrong

3.6 f 2 / 5

- **0 pts** Correct

- **2 pts** v2 is wrong

- **2 pts** state 1 not labeled

✓ - **1 pts** **v3 is wrong**

- **5 pts** not done /wrong

✓ - **2 pts** **dome missing**

- **1 pts** state 3 is wrong

- **1 pts** v1 is wrong

Roxanna Mendoza

CHE 201 Midterm Exam 2

#1.

FALSE

TRUE

FALSE

TRUE

FALSE

TRUE

TRUE

Part B:

$$Re = \frac{QD}{\nu A}$$

$Q = m^3/s$

$D = m$

$A = m^2$

$\nu = m^2/s$

$$\frac{\frac{m^3}{s} \cdot m}{m^2} \qquad \frac{\frac{m^4}{s}}{m^2 \cdot \frac{m^2}{s}}$$

1.1 A **12 / 21**

- **- 0 pts** Correct
- ✓ **- 3 pts first one is incorrect**
- **- 3 pts** second one is incorrect
- **- 3 pts** third one is incorrect
- ✓ **- 3 pts fourth one is incorrect**
- **- 3 pts** fifth one is incorrect
- **- 3 pts** sixth one is incorrect
- ✓ **- 3 pts seventh one is incorrect**

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CHE 201 Midterm Exam 2

#1. FALSE

TRUE

FALSE

TRUE

FALSE

TRUE

TRUE

Part B:

$$Re = \frac{QD}{\nu A}$$

$Q = m^3/s$

$D = m$

$A = m^2$

$\nu = m^2/s$

$$\frac{\frac{m^3}{s} \cdot m}{m^2} \qquad \frac{\frac{m^4}{s}}{m^2 \cdot \frac{m^2}{s}}$$

1.2 B **4 / 4**

✓ **- 0 pts** Correct

- 4 pts not done or incorrect

#12. 4 kg Pan & cooking & 3 times a week

$T_{avg} = 25°C$ → Average temperature of kitchen

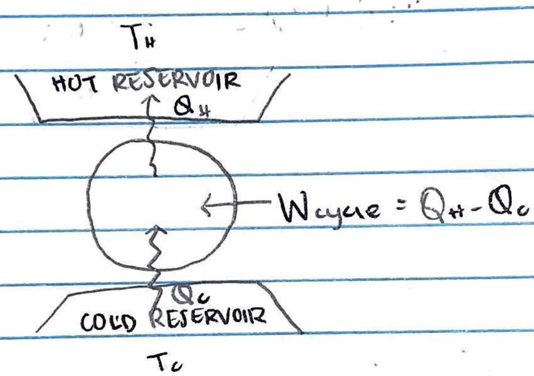
$T_{avg} = 90°C$ → Average temperature of food

$T = 4°C$ → Fridge

$C_s = 3.5\ kJ/kg\,°C$ → specific heat of food

$\beta = 1.7$

Part A:



$T_H = 90°C \rightarrow 363.15\ K$

$T_C = 4°C \rightarrow 277.15\ K$

specific heat

$Q = mc\Delta T$ ← change in Temperature

mass

$$1.7 = \frac{Q_C}{Q_H - Q_C}$$

$Q = (4\ kg)(3.5\ kJ/kg°C)(86°C)$ from Pan to Refrigerator no RT

$$\frac{3.5\ kJ}{kg°C} \cdot \frac{1000\ J}{1\ kJ} = 3500\ J$$

$Q = (4\ kg)(3500\ J/kg°C)(86°C)$ $Q = 1{,}204{,}000\ J$

52 weeks in a year

$52 \times 3 = 156$

$1{,}204{,}000\ J \times 156$

$= 187{,}824{,}000\ J$ / per year

$Q = mc\Delta T$

$(4\ kg)(3500\ J/kg\,°C)(25-4\,°C)$

$Q = 294{,}000\ J$

$\times 156$

$Q = 45864000\ J$ per year

Part B

$187824000 - 45864000 = 141{,}960{,}000\ J/year$

Part E

$$W = \frac{Q_c}{COP}$$

$$W = \frac{141{,}960{,}000\ J/yr}{1.7} \cdot \frac{1\ kJ}{1000\ J} \cdot \frac{1\ kWh}{3600\ kJ} = 23.17\ kW$$

$(0.12)(23.17) = \$2.78$

$$23.17\ kW = \frac{Q_c}{1.7}$$

$Q_c = 39.389\ kJ$

$Q_H = 12.4359\ kJ$

Part D

$$\beta_{max} = \frac{T_c}{T_H - T_c} = \frac{277.15\ K}{363.15 - 277.15} = 3.22$$

Part F

$$-\sigma = \frac{Q_{in}}{T_H} - \frac{Q_{out}}{T_C}$$

$$-\sigma = \frac{39.389}{363.15\,K} - \frac{12.4359}{277.15\,K}$$

$$-\sigma = 0.06359 \ kJ/K$$

Therefore this process is impossible

2.1 a **10 / 10**

✓ **- 0 pts** Correct

- 3 pts Qc not labeled or wrong direction

- 3 pts Qh not labeled or wrong direction

- 3 pts work not labeled or wrong direction

2.2 b **5 / 5**

✓ **- 0 pts** Correct

- 5 pts incorrect or not done

- 2 pts only one calculated

- 2 pts partially done/not calculated for year

- 2 pts without cooling is incorrect

2.3 C **0 / 5**

- **- 0 pts** Correct
- ✓ **- 5 pts** not done/incorrect
- **- 1 pts** calculation error

2.4 d **2 / 5**

- **- 0 pts** Correct
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- ✓ **- 3 pts Th is taken as 90C**

2.5 e **10 / 10**

- ✓ **- 0 pts** Correct
- **- 2 pts** calculation error
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- **- 5 pts** only one calculated
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2.6 f **0 / 5**

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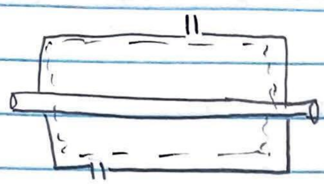
#3.



Engineering Model
① control volume
② steady state
③ $\Delta PE = 0$

$\dot{m}_{steam} = \dot{m}_{hot\ water}$

Energy rate Balance

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \sum \dot{m}_i \left(h_i + \frac{V_1^2}{2} + gz_1\right) - \sum \dot{m}_e \left(h_e + \frac{V_2^2}{2} + gz_e\right)$$

Part E: $A \cdot V$ @ inlet v @ 30 bar & T = 400 °C

$$\dot{m} = \frac{AV}{v} \qquad 2\,\frac{kg}{s} = \frac{A \cdot (40\ m/s)}{(0.0994)\ m^3/kg}$$

$$AV = 0.1988\ m^3/s$$

$\dot{m}_i = \dot{m}_{steam} + \dot{m}_{hot\ water}$

$2 = 2\dot{m}_e$

$\dot{m}_e = 1\ kg/s$

h @ P = 30 bar & T = 400 °C

$h = 3230.9\ kJ/kg$

$P_2 = P_3 = 1$ bar.

$h_2 = h_g = 2{,}675.5\ kJ/kg$

$h_3 = h_f = 417.46\ kJ/kg$

ooo ⇒ continue

$$0 = \dot{Q}_{cv} - (2\times10^5) + 2((3230.9\times10^3) + 0.5(40)^2)$$

$$- 1((2675.5\times10^3) + 0.5(100)^2) -$$

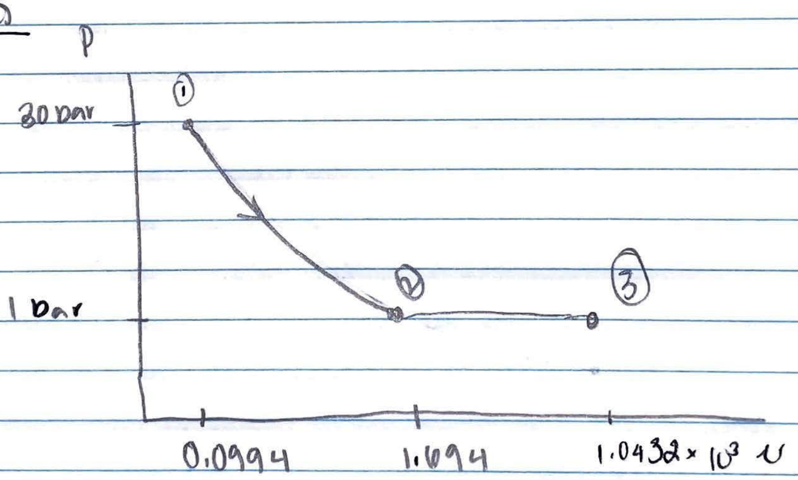
$$- 1((417.46\times10^3) + 0.5(90)^2)$$

$$0 = \dot{Q}_{cv} - (2\times10^5) + (6462600) - 2680500 - 421510$$

$$\dot{Q}_{cv} - 3560590 = 0$$

$$\dot{Q}_{cv} = -35.4 \text{ MW}$$ Part D

Part a



③ $v_f = 1.0432\times10^3 \ m^3/kg$

② $v_g = 1.694 \ m^3/kg$

① $v = 0.0994 \ m^3/kg$

3.1 a **2 / 5**

- **- 0 pts** Correct
- ✓ **- 3 pts two outlets are not shown**
- **- 2 pts** inlet is not shown
- **- 1 pts** boundary is not labeled

3.2 b **5 / 5**

✓ **- 0 pts** Correct

- 2 pts open system not stated

- 2 pts steady state is not stated

- 1 pts delta PE is zero

3.3 C 2 / 5

- **- 0 pts** Correct
- ✓ **- 5 pts incorrect**
- **- 2 pts** work can not be canceled
- **+ 2 Point adjustment**

3.4 d **10 / 10**

✓ **- 0 pts** Correct

- 2 pts h2 is wrong

- 2 pts KE term is not divided by 1000 (unit conversion)

- 10 pts incorrect or not done

- 2 pts not multiplied with mass outlets

- 1 pts work sign is wrong

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- 2 pts mass is wrong

- 2 pts h1 is wrong

- 2 pts h3 is wrong

- 0 pts Click here to replace this description.

calc error

3.5 e **5 / 5**

✓ **- 0 pts** Correct

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- 2 pts specific volume is wrong

3.6 f **2 / 5**

- **- 0 pts** Correct
- **- 2 pts** v2 is wrong
- **- 2 pts** state 1 not labeled
- ✓ **- 1 pts** **v3 is wrong**
- **- 5 pts** not done /wrong
- ✓ **- 2 pts** **dome missing**
- **- 1 pts** state 3 is wrong
- **- 1 pts** v1 is wrong